Effects of Imipramine Treatment of Separation-Induced Social Disorders in Rhesus Monkeys

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• **Two groups of young rhesus monkeys were subjected to repetitive peer separations, a procedure that has been shown to produce depressivelike reactions in infant monkeys. Midway through the procedure one group was treated with the antidepressant imipramine hydrochloride, the other with a saline placebo. In comparison with placebo treatment, the imipramine treatment yielded significant behavioral improvement in a form and with a time course similar to that seen when the drug is given clinically to human depressives. We discuss the implications of the findings.**

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Once the object of considerable criticism and contempt,[1] animal models of human psychopathology have attracted favorable interest in recent years.[2-4] A prime reason for the rise in respectability has been the development of objective criteria by which each model can be evaluated with respect to validity and use. These criteria include similarity between etiologies of the animal model and the human disorder, similarity between behavioral manifestations, and similarity between therapeutic procedures that are effective in reversing or arresting the disorder.[5,6] The more closely the animal findings satisfy these criteria, the more acceptable and useful the model will be for the theorist and therapist alike.

One animal model that has received extensive research attention in the past decade has been that of separation-induced "depression" in monkeys. Several investigators have reported striking similarities between the reaction of some macaque infants to physical separation from mothers and the syndrome of anaclitic depression diagnosed in some human infants physically separated from their parents.[7-11] More specifically, there exist data that show that among both humans and monkeys the same etiology—maternal separation—can produce the same symptoms—agitation followed by withdrawal and inactivity, which are usually remedied by the same procedure, reunion between mother and infant. In addition, it has been found that severing of nonmaternal social bonds, eg, those between peers, can produce similar reactions among monkeys over a wide age range.[12-14] Other data suggest that monkeys exposed to social stresses early in life may be predisposed to depressive reactions to social separations later in life, in contrast to monkeys who have had a stress-free infancy and childhood.[15] As a result of these findings, a number of researchers have argued that separation in monkeys represents a valid and useful model of human depression.[8,16]

With respect to the criteria outlined above, it is apparent that separation manipulations in monkeys parallel social losses predisposing to depression among some (but certainly not all) human adolescents and adults. Moreover, certain behavioral and neurophysiological concommitants of separation in monkeys strongly resemble symptoms commonly found in some depressed human patients.[17] Unfortunately, there exists almost no data evaluating the efficacy of acceptable human antidepressant treatments for separation-induced disorders in monkeys. Without such data, the monkey separation model of depression cannot fully satisfy all criteria.

In view of these facts, the present experiment was

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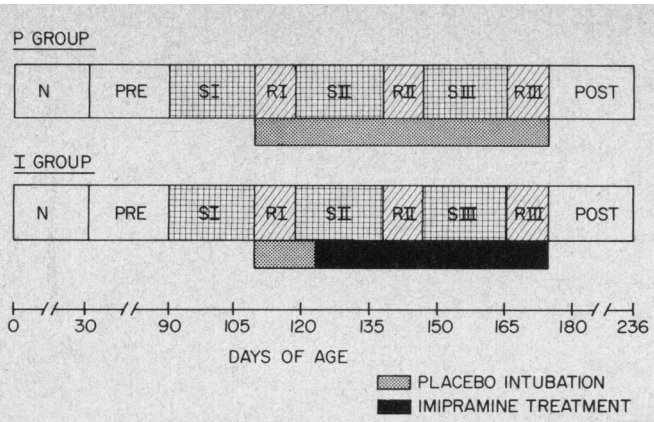


Fig 1.—Experimental design of the study. P group and I group refer to placebo monkeys and experimental monkeys, respectively; N indicates nursery housing; PRE indicates preseparation phase, SI, SII, and SIII indicate the three 19-day periods of separation, respectively; RI, RII, and RIII indicate the three 9-day periods of reunion, respectively; POST indicates postseparation phase.

designed to examine the effects of a tricyclic antidepressant in rhesus monkeys subjected to social separation. The drug imipramine hydrochloride was chosen because its clinical effectiveness as a human antidepressant is well-established, facilitating cross-species comparisons of its efficacy. An experimental paradigm involving repetitive peer separation was chosen to simulate depressive reactions in young rhesus monkeys because previous work with this species had indicated that peer separation paradigms minimized several methodological difficulties inherent in the more "traditional" mother-infant separation manipulations.[12,18]

SUBJECTS AND METHODS

The subjects were eight rhesus monkeys (*Macaca mulatta*), each removed from its mother at birth and reared for the first 30 days of life according to standard laboratory nursery procedures.[19] Two male and two female subjects, all born within a 45-day period, were assigned to the placebo (P) group; four female subjects, all born within a 26-day period, were assigned to the imipramine (I) group. Because order of the date of birth was used to determine group assignment, it was not possible to balance sexes of subjects between groups.

Rearing, Separation, and Reunion Procedures

The experimental design of the study is diagrammed in Fig 1. When the youngest monkey in each group of four reached 30 days of age, the group was removed from the laboratory nursery and placed in a 1.52 × 1.22 × 0.76-m living cage, giving group members unlimited physical access to each other 24 hours per day. Such group housing early in life has been demonstrated to yield strong mutual attachments in infant monkeys.[12,20] Each group remained together for eight weeks prior to the initial separation (preseparation phase).

When the age of the youngest member of each group reached 90 days, the separation phase of the study was initiated. Each member of the group was removed from the living cage and housed individually in a 0.76 × 0.61 × 0.76-m bare, wire-mesh cage in another room, where it could see and hear but not physically contact other group members. After 19 days of such separation, all subjects were returned to their living cage for a nine-day period of reunion. Immediately after the ninth day of reunion, the subjects were separated again for a second 19-day period, housed exactly as they had been during the first separation. The monkeys were then reunited for nine days, in the same manner as before. A third 19-day separation and nine-day reunion followed.

After the final day of the third reunion, the group members were permitted unlimited mutual interaction 24 hours per day without interruption, as was the case prior to the first separation. They were maintained in this manner for an additional 12-week period (postseparation phase).

Drug/Placebo Administration Procedure

Beginning on the first day of the first reunion, placebo intubation was initiated for all subjects in both groups. Placebo intubation consisted of administration of 1 ml/kg of body weight tap water via nasogastric intubation. It was performed between 2 PM and 3 PM daily, seven days per week.

On the eighth day of the second separation, after 16 days of placebo intubation, actual drug treatment was begun for members of the I group, and it was continued on a daily basis until the end of the third reunion period. The particular design for placebo and drug treatment schedules was chosen to permit both between-group and within-group comparisons for evaluation of placebo tubing and actual treatment effects. Placebo intubation was begun in all groups on the first day of the first reunion and continued into the second week so that for each group a comparison of tubing and nontubing in two contexts, both in group housing and in separation housing conditions, could be made. Since previous work had shown that repeated separations yield similar behavioral reactions,[12] it was believed that the confounding with separation order would have minimal effects. Further, the anticipated absence of order effects would enhance evaluation of drug-placebo differences within individual subjects as well as between subject groups.

Each I monkey was treated with a solution of imipramine hydrochloride dissolved in water. Dosage was set at 10 mg/kg of body weight. The dosage level was chosen on the basis of pilot testing for effectiveness in reversing reserpine-induced sedation. The concentration of the solution was 10 mg/ml, allowing the dosage volume to remain at 1 ml/kg, equivalent to that for placebo tubing. The P-group monkeys continued to receive placebo intubations as before.

To summarize, during the first separation no intubation or drug administration took place. Placebo intubation began on the first day of the first reunion, and actual administration of imipramine began on the eighth day of the second separation and continued through the final day of the third reunion. There was no intubation or drug administration during the postseparation phase of the experiment.

Data Collection and Analysis

Beginning at 30 days of age, each subject was observed for at least one 7.5-minute session five days each week, and these observations were continued through the end of the study. During each observation session, frequency and duration of behaviors falling into each of 13 predefined behavioral categories were recorded on a set of clocks and counters by one of three experienced testers. Behaviors encompassed by each category are presented below; they have been employed in previous studies of monkey separation.[12,21]

Ventral cling: Contact of own ventral body surface with another subject.

Self-clasp: Clasping of any part of own body with hand(s) and/or foot (feet).

Self-mouth: Oral contact with own body.

Environmental exploration: Tactual and/or oral manipulation by subject of inanimate objects.

Rock and huddle: Huddle accompanied by stereotypic rocking.

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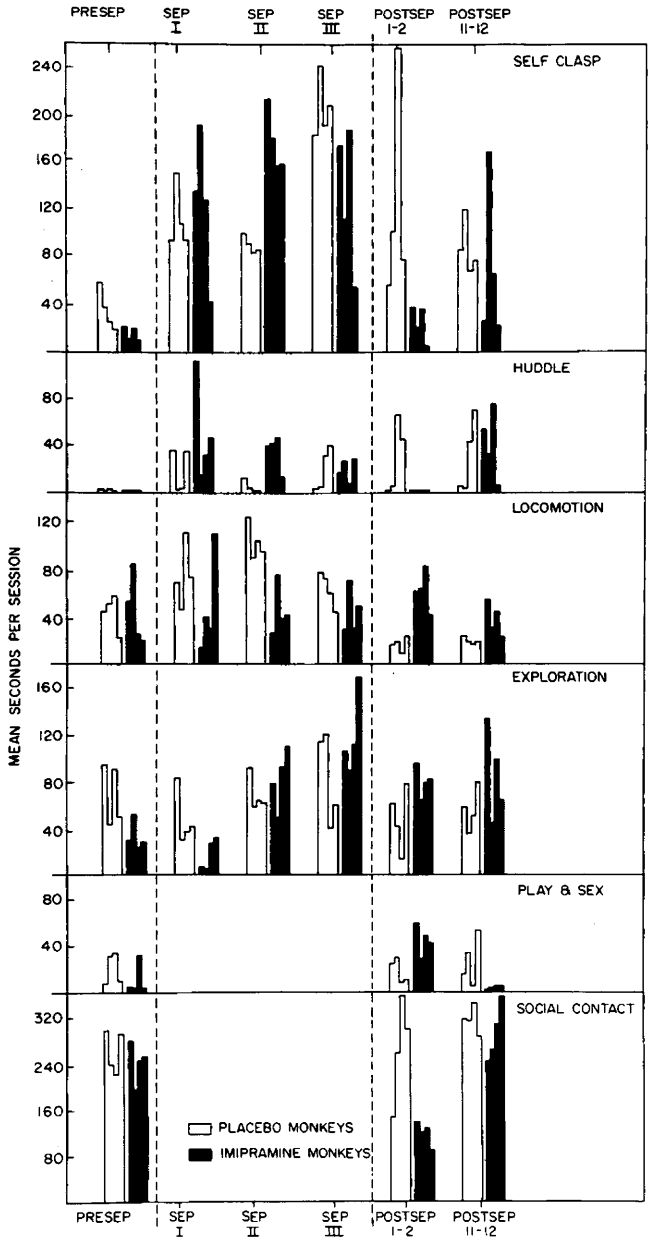


Fig 2.—Mean levels for each subject for six behavioral categories (self clasp, huddle, locomotion, exploration, play and sex, and social contact) during each of six time periods (Preseparation, Separations I, II, and III, and Postseparation 1-2 and 11-12). For members of the P Group, the first two scores for each period are those of the two male subjects; last two are those of the two female subjects (all I group members were female).

Aggression: Vigorous and/or prolonged biting, hair pulling, clasping, accompanied by one or more of threat, barking, piloerection, or strutting.

Passive: Absence of all social, exploratory, and locomotor activity.

Stereotypy: Patterned movements maintained in a rhythmic and repetitive fashion for at least three cycles (scored independently and noninclusive of locomotion).

Locomotion: Any self-induced change in location of self.

Social contact: Contact and/or proximity (within five centimeters) with another subject.

Play and sex: Any sexual advances or posturing directed toward another subject and/or any type of play activity (including contact and noncontact play).

Vocalization: Any sound emitted by subject.

Huddle: Self-enclosed, fetal-like position.

The testers had been trained to a rigorous laboratory criterion of interobserver reliability ($r \geq +.95$) immediately prior to the beginning of the study. Each tester remained blind to the drug status and schedule of each group until the study had been completed. All testing was finished before 1:00 PM each day, well before intubation commenced on the days when drugs were administered to the subjects.

The data contained in the present report consisted of the mean duration scores for each behavioral category calculated for every subject over each of six time periods: the last 14 days of preseparation, each of the three 19-day periods of separation, and two 14-day periods during postseparation (the first and the last, respectively). These time periods were chosen to permit comparisons between I and P monkeys (*a*) immediately prior to separation, (*b*) during separation but prior to drug or placebo treatment, (*c*) during separation in the initial stages of treatment, (*d*) during separation when treatment was chronic, (*e*) immediately after termination of both treatment and separation, and (*f*) in a follow-up observation ten weeks later. Statistical comparisons between groups were made for each of the six time periods, using nonparametric Mann-Whitney tests.[22] Statistical significance was a priori set at $P \leq .029$ (1-tail test)—the probability that there would be no more than a single overlap between two sets of four scores each.

RESULTS

Despite comparable levels of behaviors prior to the separation and similar reactions to the first separation, monkeys in the two groups exhibited different behavioral trends once treatment was initiated. By the end of the treatment period, imipramine-treated subjects appeared less severely affected by the separation procedure than their placebo-treated controls. Differences between groups diminished gradually after treatment was discontinued. These findings are summarized in Fig 2, which presents for every subject the mean score for several behavioral categories during each of the six time periods described above. Details of the findings for each period follow.

Preseparation

Analysis of behavioral scores recorded during the last two weeks of preseparation revealed no statistically significant differences between P and I monkeys for any of the 13 behavioral categories. In general, the scores exhibited by these subjects were comparable to those shown by 3-month-old monkeys reared in similar social environments in previous studies.[23-26] Moreover, sex differences in behavioral levels within the sex-balanced P group were consistent with those reported in the developmental literature,[27] while female P-group members' levels for each behavioral category overlapped those of the all-female I group (Fig 2). Thus, the data indicated that prior to separation, all subjects in both groups were displaying behavioral levels appropriate for rhesus monkeys of their own age, sex, and rearing history.

Separation I (No Treatment)

Both P and I groups responded to the first separation in a manner consistent with that reported in previous studies

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of peer separation in infant rhesus monkeys.[12,16,21] As shown in Fig 2, monkeys in both groups exhibited significant increases over preseparation levels in selfclasping and huddling behaviors, nonsignificant increases in locomotion, and nonsignificant decreases in exploration. In addition, they both displayed marked increases in vocalization and stereotypy over preseparation levels. (We also have performed much more complete analyses, using all the data collected on these subjects during the entire study. The analyses involved repeated-measures analysis of variance with groups as an independent variable and various time periods as repeated measures. The results confirmed all the findings of the present report, in addition to revealing numerous significant time effects and group X time interactions. While discussion of the latter findings are beyond the scope of the present report, results of all the analyses may be obtained from us on request.) Group differences during separation 1 were found in only one behavioral category—exploration. Here, P monkeys displayed higher levels than I monkeys (one-tail $P = .029$). In all other categories of behavior, P and I monkeys were similar in their reaction to peer separation prior to treatment.

Separation II (Treatment Begins)

In contrast to the first separation, separation II was marked by several differences between P and I monkeys. These included higher levels of self-clasp and huddle by I monkeys (both Ps = .014) and higher levels of locomotion by P monkeys ($P = .014$). In addition, I monkeys showed an increase in levels of exploration over those exhibited in Separation I, whereas P monkeys displayed no such increases. Such differences between groups were not due to the differential sex composition of the two groups, for on all of the above behaviors P-group female subjects did not overlap with any of the scores from the all-female I group (Fig 2). Instead, it seems likely that the group differences could be attributed to the different treatments received by I and P monkeys during separation II. However, the data indicated that the immediate effects of initiating imipramine treatment could hardly be considered to be antidepressant in nature. If anything, the increased levels of self-clasp and huddle and decreased level of locomotion for imipramine-treated monkeys suggested that their depressive-like behaviors were becoming more pronounced during the second separation.

Separation III (Chronic Treatment)

The group differences observed during the second separation were largely reversed in separation III. Most strikingly, self-clasping by I monkeys was less than that by P monkeys ($P = .029$) during the third separation; exactly the opposite had been true during separation II. The other behaviors that differentiated the two groups in the second separation failed to do so in the third. In fact, inspection of the data indicated that during separation III, the female subjects in the P group displayed higher levels of huddle and lower levels of exploration than any members of the all-female I group (Fig 2). These findings suggest that chronic imipramine treatment was associated with less passive self-directed behavior (self-clasp, huddle) and more interest in the environment (exploration) in comparison with same-sex placebo-treated controls. From a different perspective, Fig 2 shows that after chronic imipramine treatment, monkeys reacted to separation with generally less self-directed behavior and more environmentally-directed activity than they did at the start of treatment, while after chronic placebo treatment, monkeys reacted to separation with more self-directed behavior and less environmentally-directed activity than they did when placebo treatment was begun.

Postseparation: First Two Weeks (Treatment Ends)

The P and I monkeys differed from each other in more levels of behavior during the first two weeks of postseparation than during any other time period examined. The I monkeys displayed higher levels of locomotion ($P = .014$) and of exploration, play, and sex (Ps = .029); P monkeys exhibited higher levels of self-clasp and social contact (Ps = .014), as well as huddle ($P = .029$). These findings are significant in light of previous peer separation data, which have consistently shown that such separations have cumulative effects of increasing self-directed behavior over that shown prior to separation, as well as severely arresting development of age-appropriate social behavior.[12,21,25] The P monkeys were consistent with those in previous studies, in that their levels of self-clasping and huddling were higher after the separations than they were in preseparation, while their social behavioral development was arrested, that is they continued to display high levels of infantile contact-maintaining behavior and low levels of age-appropriate play and sex activities. In contrast, I monkeys displayed low levels of self-directed and contact-maintaining behaviors and elevated levels of exploration, play, and sex. Their behavioral repertoires following separation were more similar to those reported in the literature for like-aged monkeys reared together but never separated.[20,21,23-25]

Postseparation: Last Two Weeks (Follow-up)

Data collected during the last two weeks of postseparation revealed that 11 to 12 weeks after cessation of imipramine administration most treatment effects had disappeared. Only for the category of locomotion did I subjects maintain the advantage over P monkeys that they had held at the start of postseparation. For one category, play and sex, the advantage shifted to P monkeys, although the difference between groups ($P = .028$) may have been in part due to differences in sex composition between groups. On all other behaviors there were no significant differences between P and I monkeys during this follow-up period.

COMMENT

The results of the present experiment suggested that imipramine treatment influenced the behavior of monkeys during and immediately following peer separation, but that such effects were largely transitory once treatment had ceased. These findings are entirely consistent with the predictions of a hypothetically valid animal model of depression with respect to imipramine effects.[2] Such a model would require that imipramine alleviate some of the

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symptoms of the subjects, as it does for numerous depressed human patients. It would also require that treatment have a two to three week latency for therapeutic effectiveness, as it does in most human cases. Once treatment stopped it would be expected that beneficial effects of the imipramine would gradually decrease, consistent with the human data. In all respects, the present findings meet these standards.

It must be pointed out, however, that these data by no means fully satisfy the therapy criterion for a valid model of human depression. At the very least one would have to demonstrate that the therapeutic effects seen in the monkey subjects were specific to imipramine and other antidepressants, rather than to other active agents (eg, valium, thorazine, or aspirin). Further, one would have to show that imipramine treatment did not have comparable effects for monkeys exhibiting other psychopathological syndromes, eg, that resulting from social isolation initiated at birth, or for monkeys exhibiting relatively normal patterns of behavior. These issues are fortunately quite amenable to experimental scrutiny, and so they are now basically empirical in nature.

If additional research does show that the monkey peer separation model of depression is indeed valid with respect to the therapy criterion, then the model has considerable potential use. For example, it could be used to identify mechanisms underlying the actions of specific therapeutic treatments, a research approach that would be experimentally difficult and ethically questionable with human patients. At the very least, such a model has the potential to provide a far more appropriate method for screening potential human antidepressants prior to actual clinical trials than presently exists.

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Ms Dady Blake and Ms Polly Fiedler assisted in the data collection and reduction, Ms Helen LeRoy edited the manuscript, and Ms Terri Schwoegler typed the final draft.

Nonproprietary Name and Trademarks of Drug

Imipramine hydrochloride—*Presamine, Tofranil.*

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